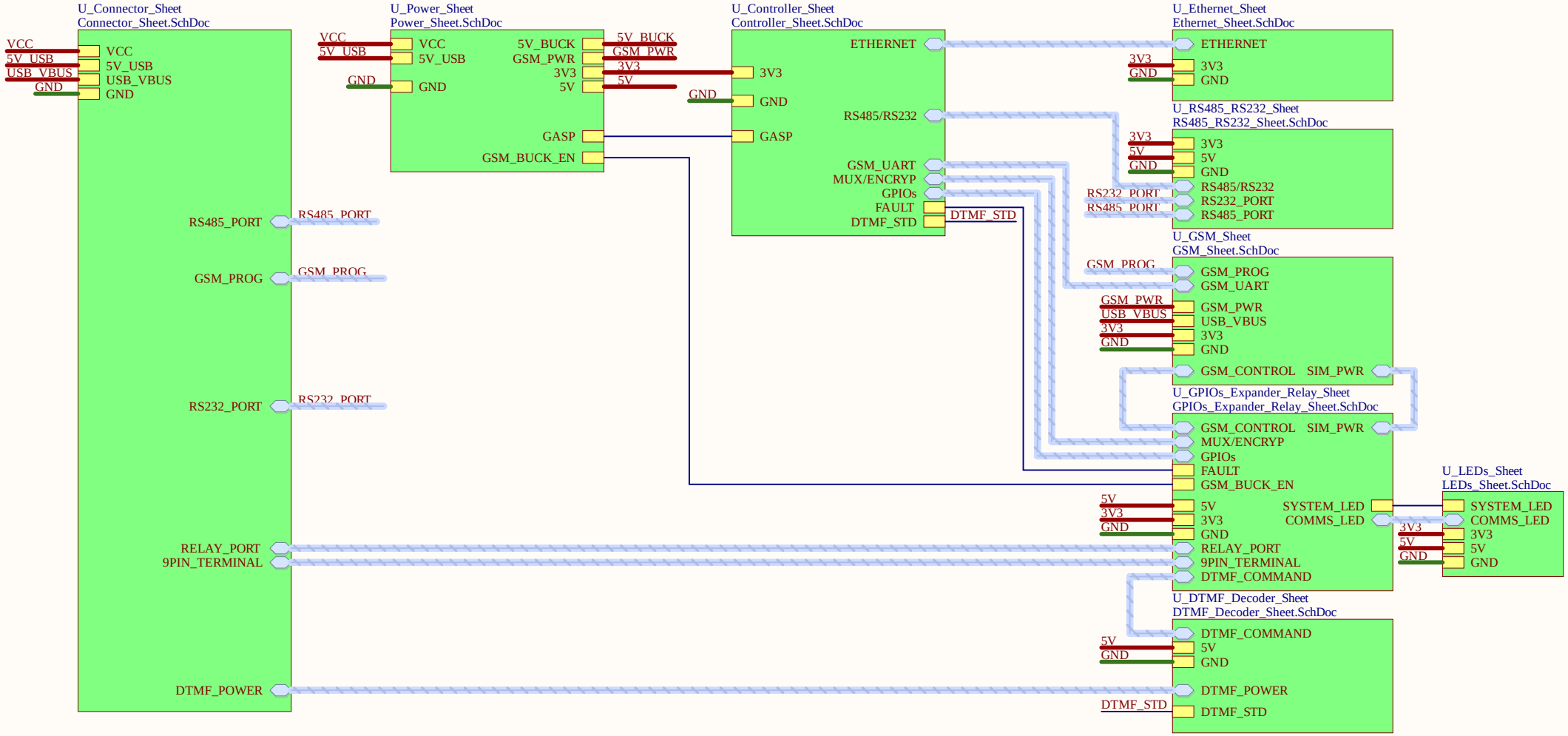
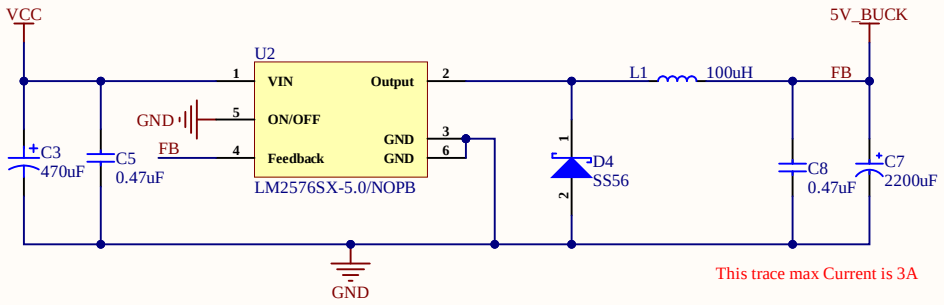


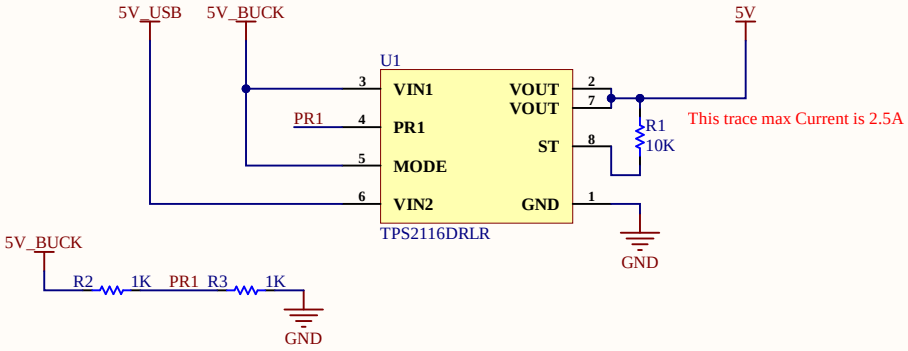
eSpark Consulting Group

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Date: 03/05/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 10

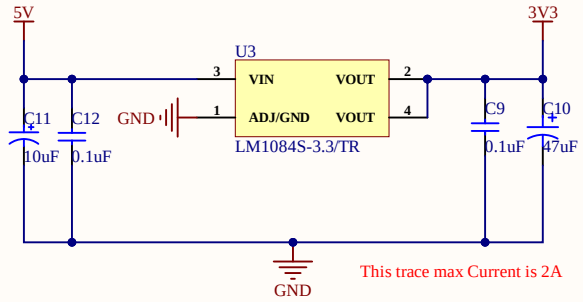
5V BUCK REGULATOR



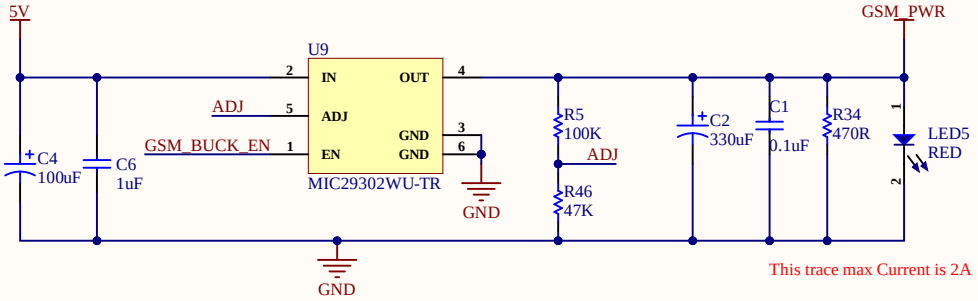
ORING CIRCUIT



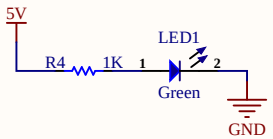
LDO 3.3V



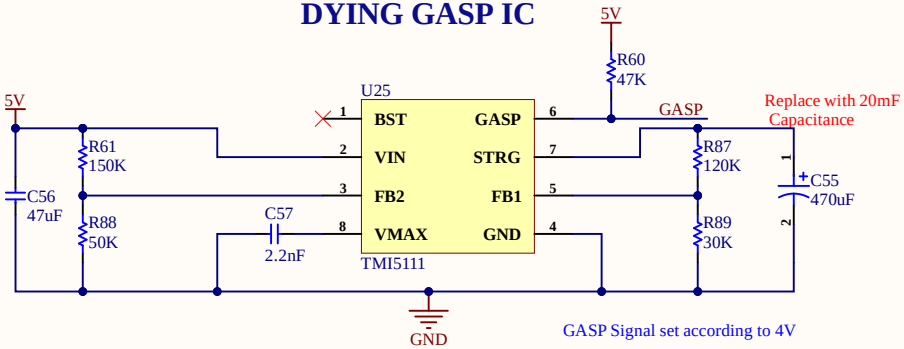
Buck for GSM Power

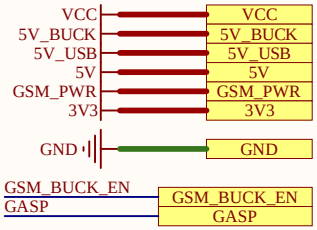


POWER LED



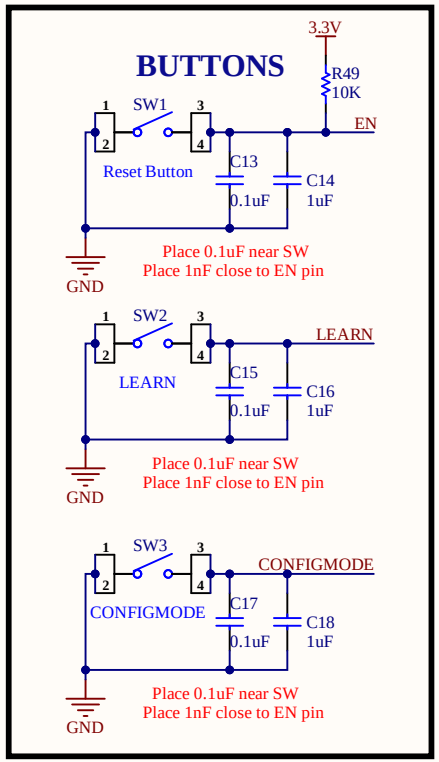
DYING GASP IC

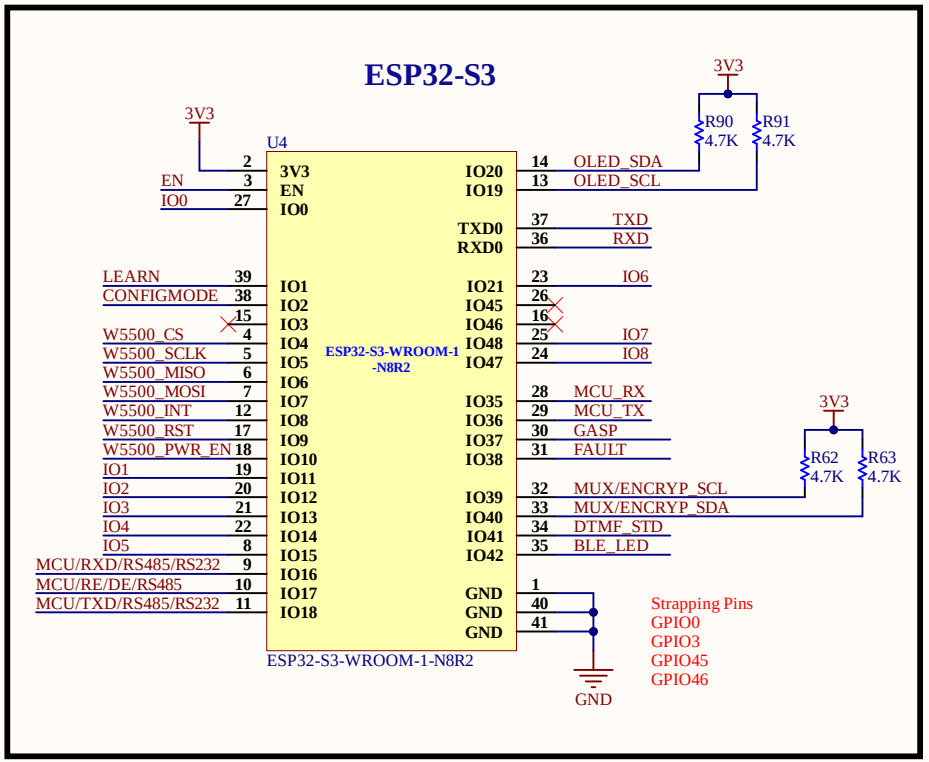


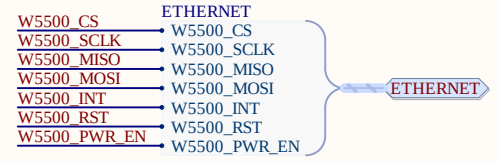


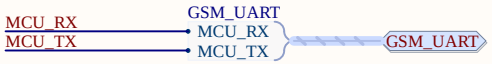
eSpark Consulting Group

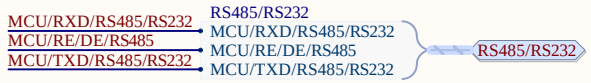
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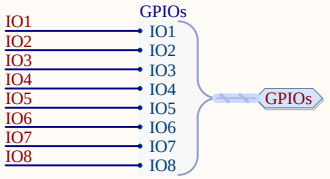




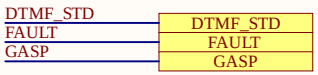


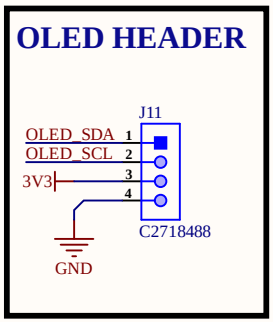


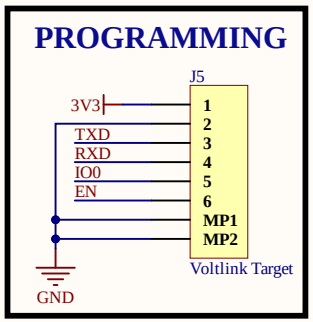


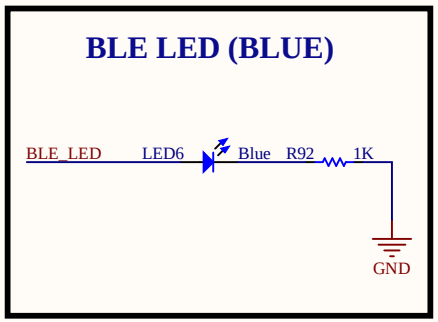


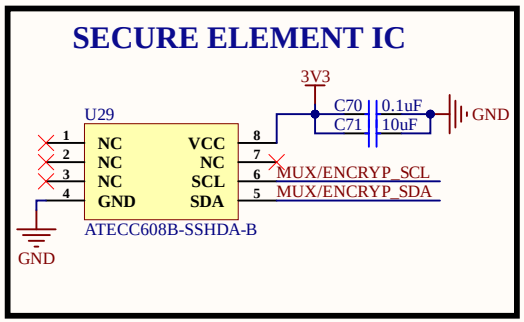








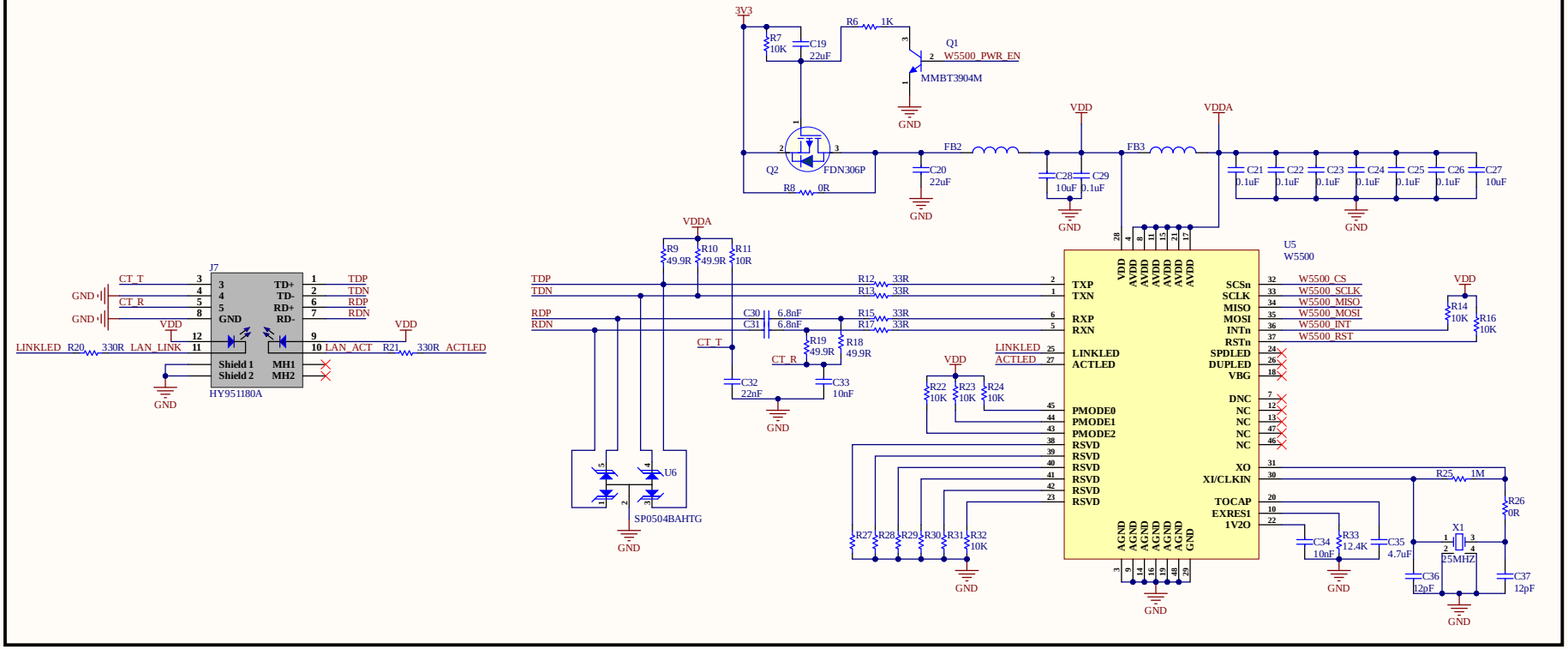




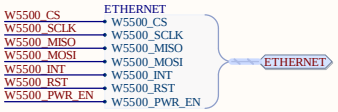


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Date: 03/05/2023	Page	Num: 3
Drawn By: Huzaifa Ali	Page	Total: 10

ETHERNET/ W5500 PHY



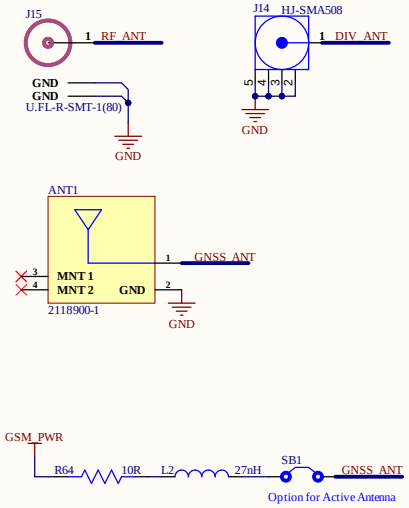




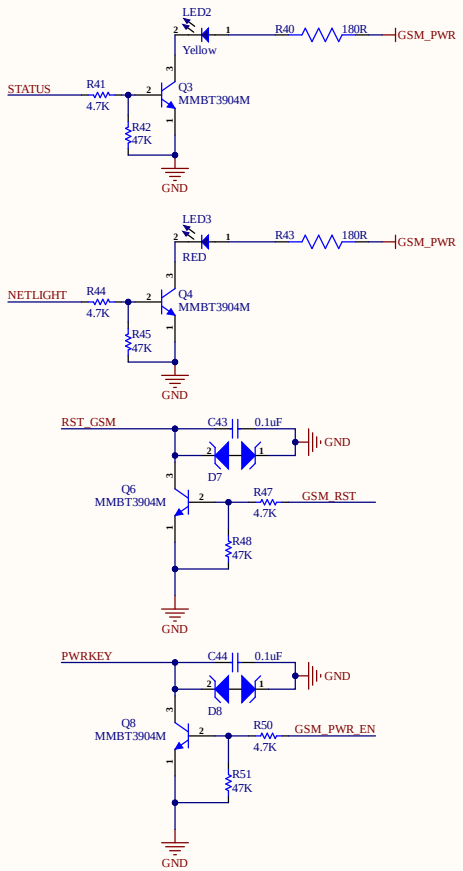
eSpark Consulting Group

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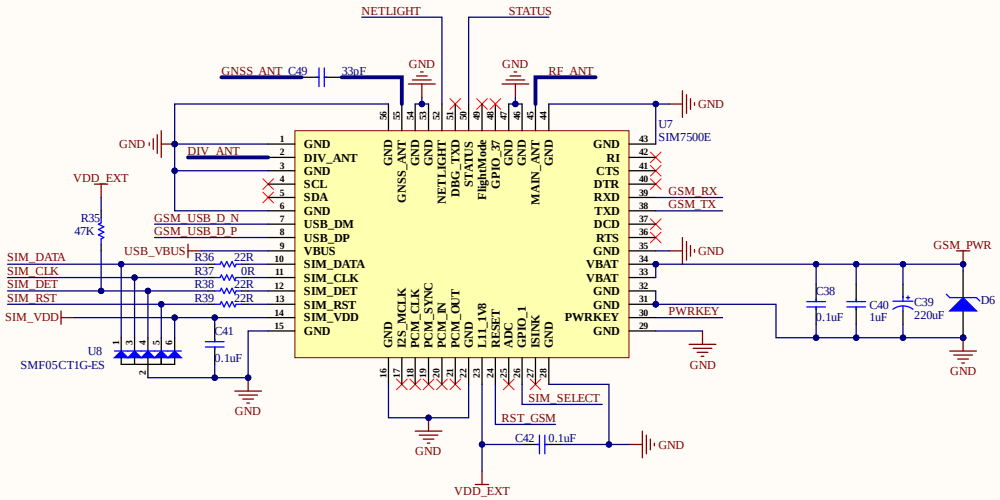
ANTENNA



GSM_COMMANDS



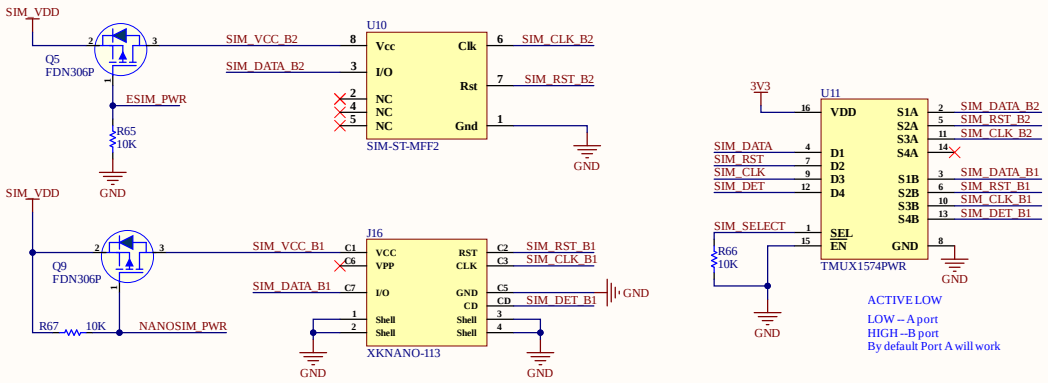
SIM7500E



SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

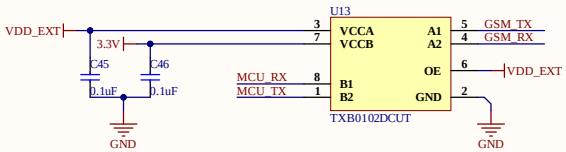
Datasheet Pg 31

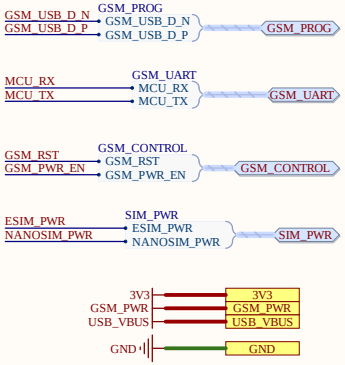
DUALSIM WITH MULTIPLEXER



ACTIVE LOW
LOW -- A port
HIGH --B port
By default Port A will work

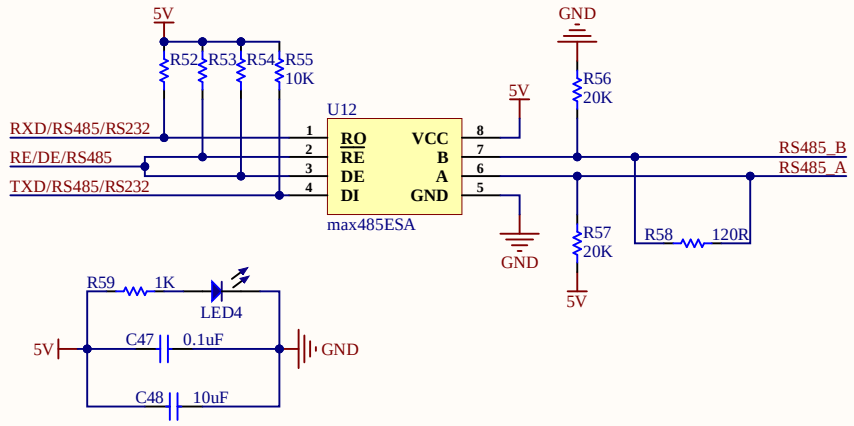
Level Shifter for GSM Serial



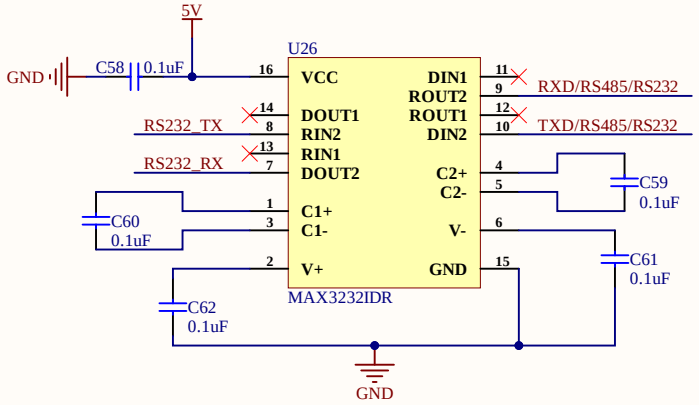


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Date: 03/05/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 10

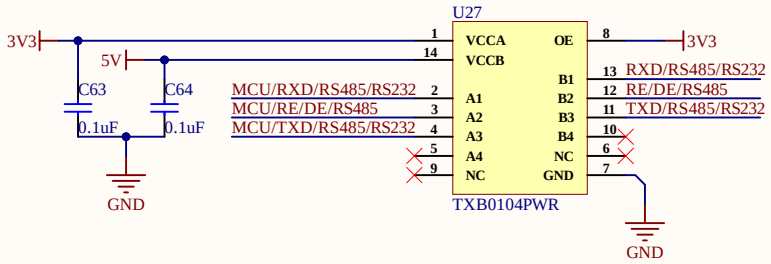
RS485 to UART

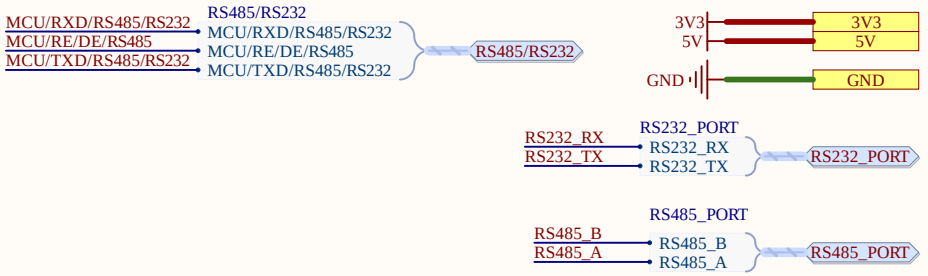


RS232 to UART



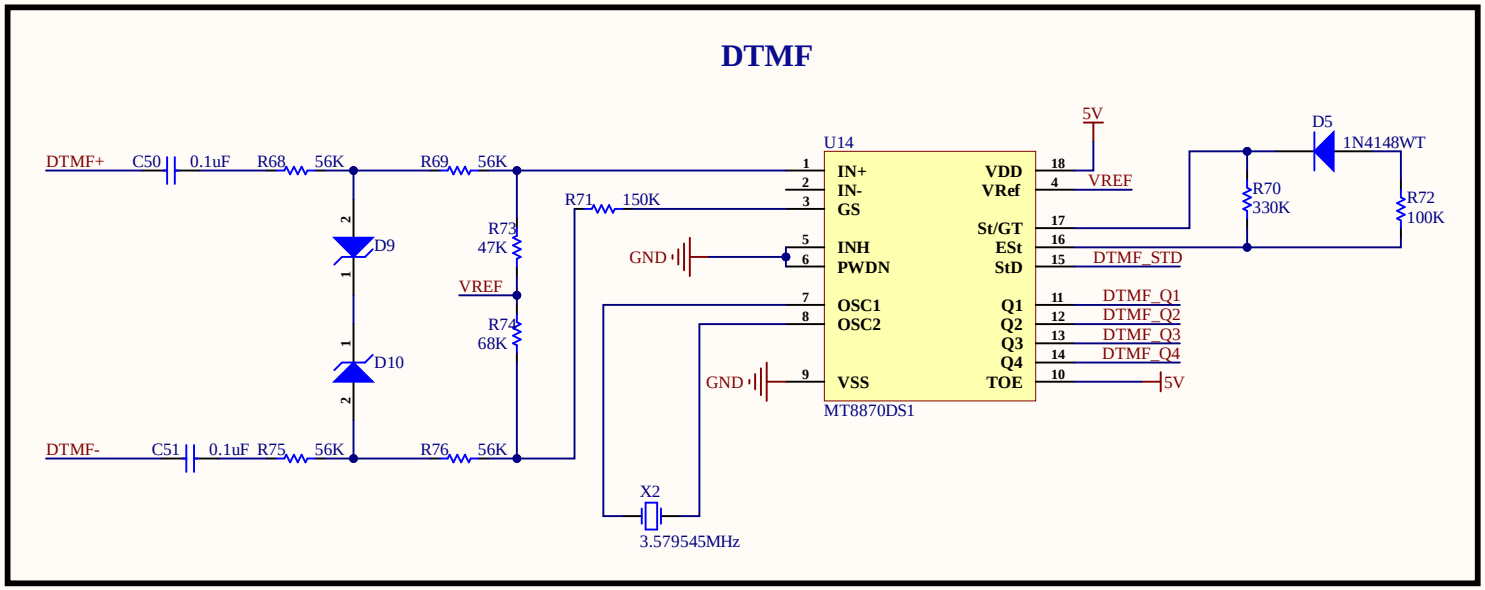
Level Shifter

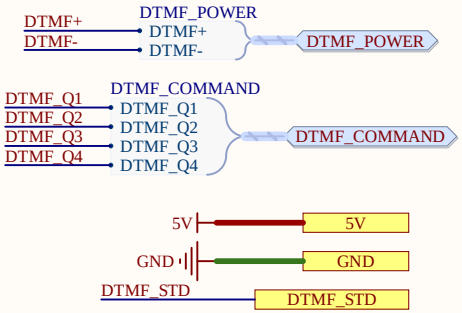




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Date: 03/05/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 10

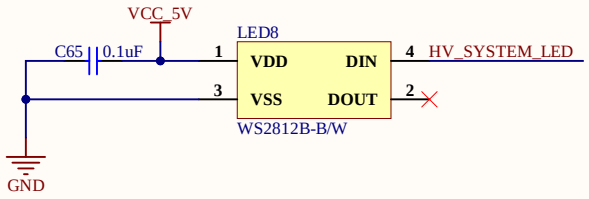




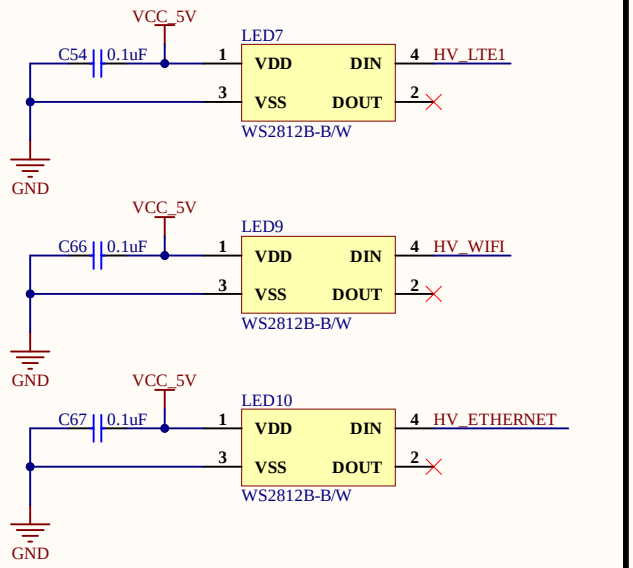
eSpark Consulting Group

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Date: 02/05/2023	Page Num: 7
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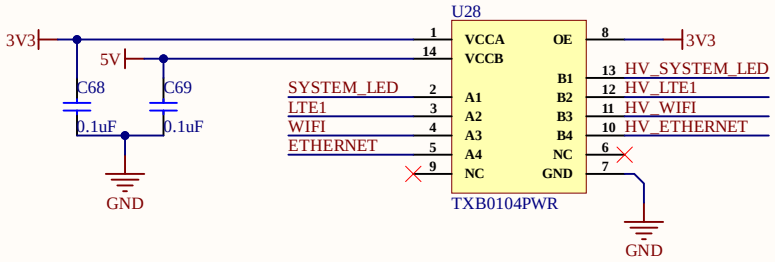
SYSTEM LED

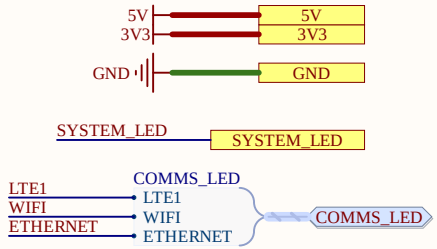


COMMS LED

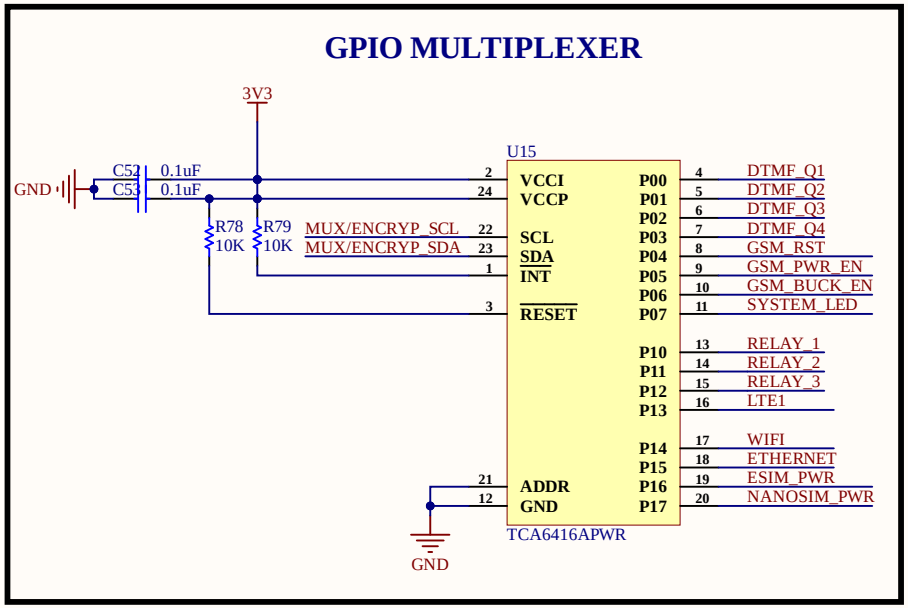


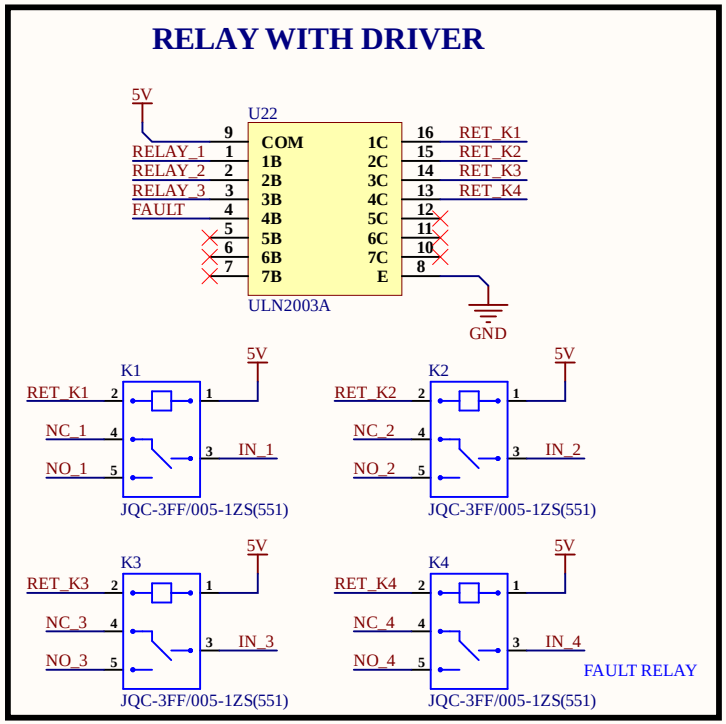
Level Shifter

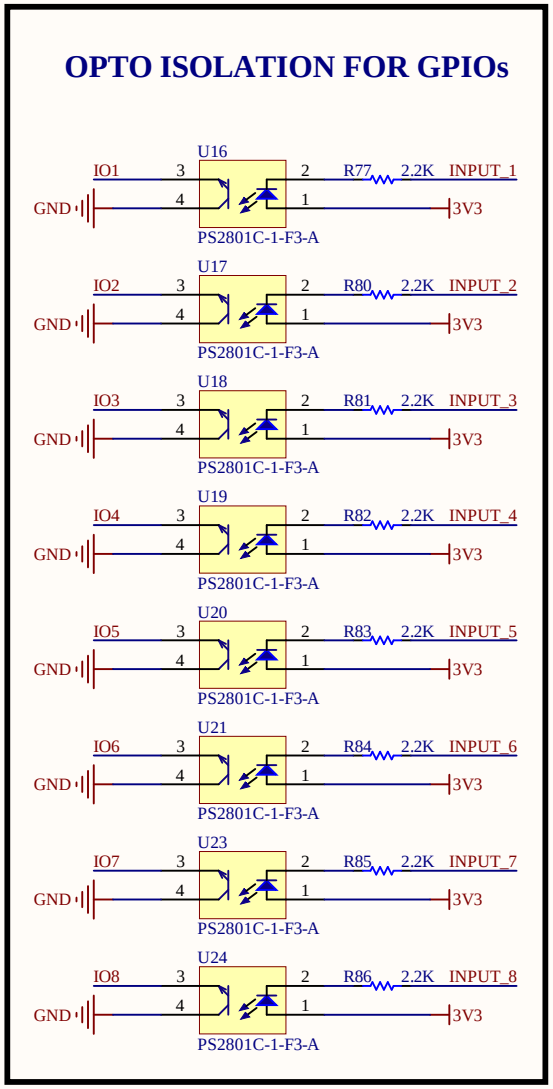


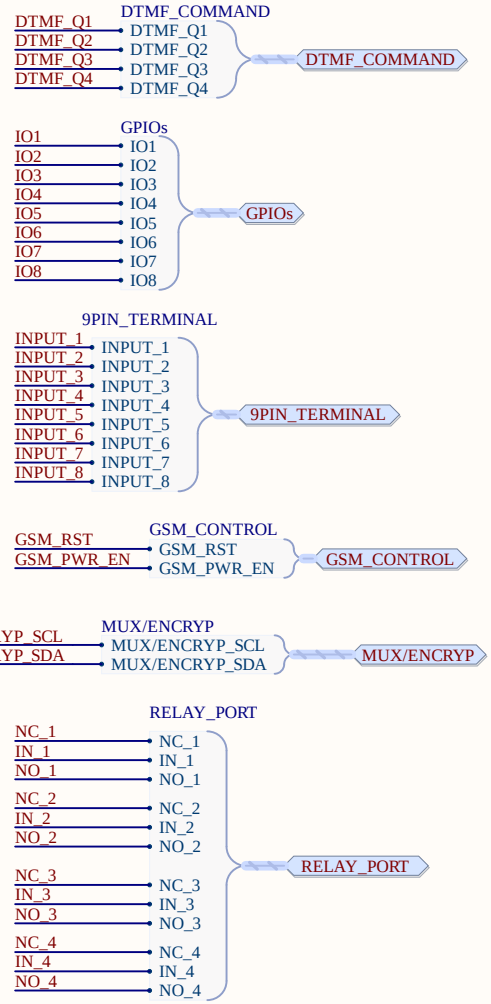


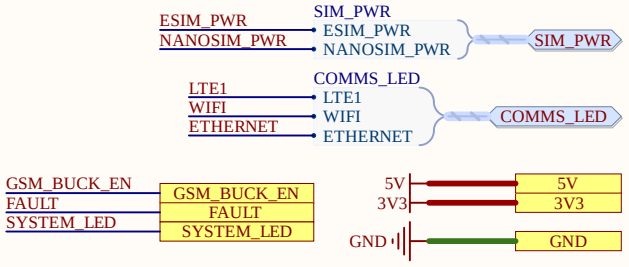
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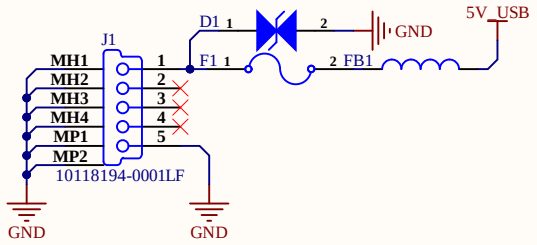




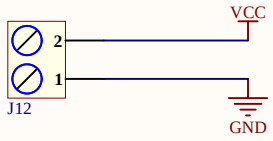


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Date: 03/05/2023	Page Num: 9
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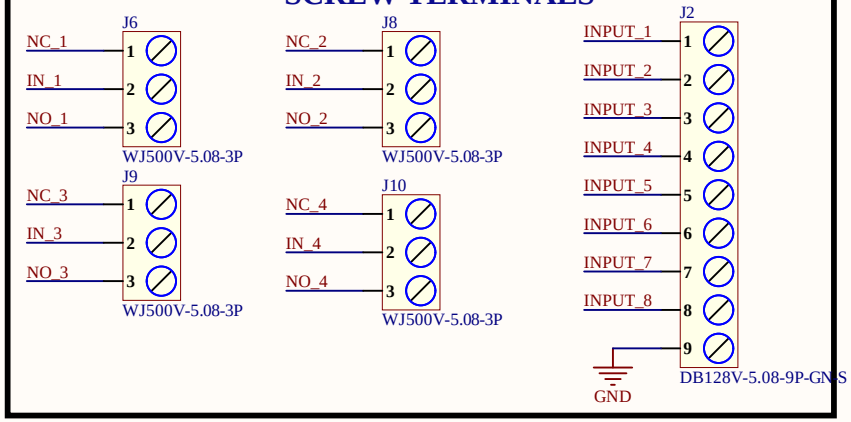
USB to Power Board



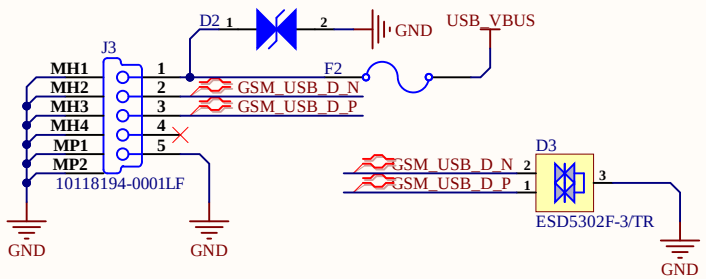
9-30V DC INPUT



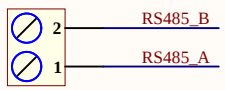
SCREW TERMINALS



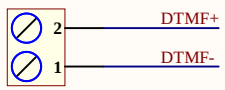
USB for GSM Updates



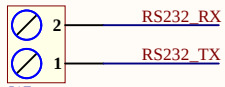
PORT for RS485

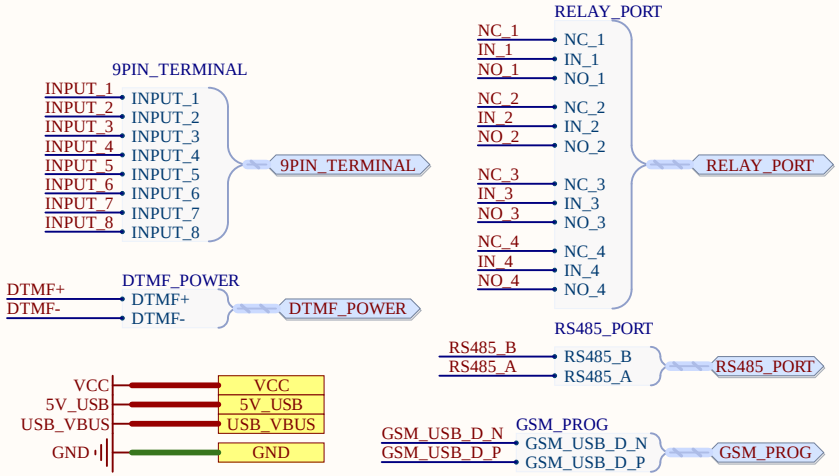


PORT for DTMF



PORT for RS232





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Title: Connector_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
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